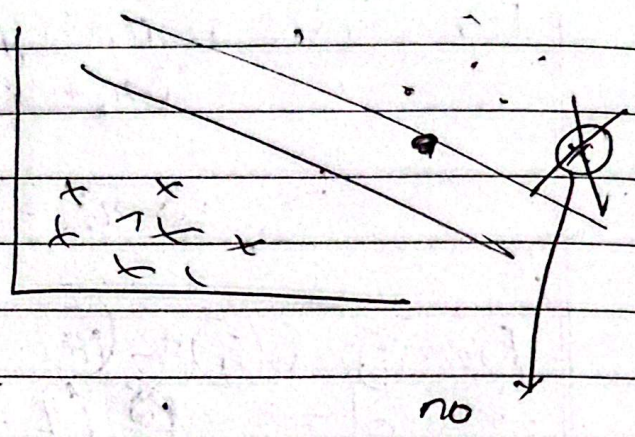
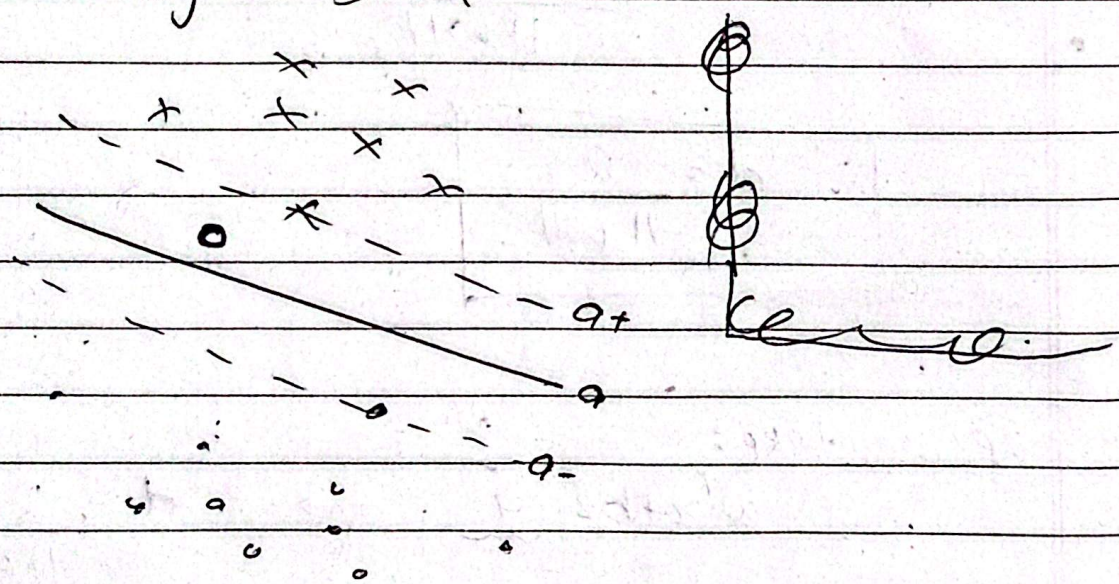
Hard margin SVM



Clearly with no errors.
-ve place can't have

Need totally clean data.

Soft Margin SVM.



will have : $y(\vec{w}.\vec{x_i} + b) \geq 1$

-1 $\quad$ $+ve$

Hence:

$y(\vec{w}\vec{x} + b) \leq -1$

so, not work

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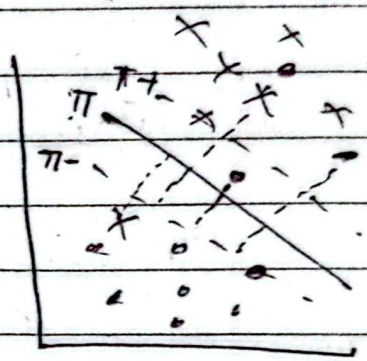
$$\max f(x) \Leftrightarrow \min \frac{1}{f(x)}$$

zeta

- SVM $\underset{(w^*, b^*)}{\arg\min}$: $\underbrace{\frac{\|w\|}{2}}_{\text{loss fn}} + \sum_{i=1}^{n} \zeta_i$



- For correcly classified points: $\zeta_i = 0$
- But, incorrectly classified points, will have certain value of ζ_i
- ζ_i for incorrect • : distance betn that point & π−
- ζ_i for correct X : distance betn that point & π+
- we want to minimize it:

SVM $\underset{(w^*, b^*)}{\arg\min}$: $\underbrace{\frac{\|w\|}{2}}_{\text{marign error}} + \underbrace{C \sum_{i=1}^{n} \zeta_i}_{\text{classification error}}$

↓ higher margin → loss margin error

↓ to reduce it we multiply by C, which is hyperparameter.